

Control of adaptation: steering an adaptive body with a designed adaptive implant

First report on the co-evolution project

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Abstract

We consider two adaptive agents, a body and an implant, that adapt to each other until they reach an equilibrium. The designer can neither act on the body nor prescribe the implant’s actions, but designs the implant’s adaptation mechanism, here its objective. The quality of a design is judged by the equilibrium that the coupled system reaches: how close the body ends to a desired healthy state and how well the implant is integrated. We position this *control of adaptation* against the two-learner problem in neural interfaces, co-adaptation, human-in-the-loop optimization, adaptive and dual control, learning in games, and strategic delegation. A minimal two-player differential game with one state, one control, and one objective per player, solved with the optimal-control based solver CorleoneGame.jl, shows a non-trivial effect: an implant that pursues exactly the designer’s objective is exploited, because the body lets the implant do the adapting. An implant designed to disregard the mismatch leads the body to the healthy state and yields a markedly lower designer cost, although the designer values integration. This is the commitment effect known from strategic delegation, here in a dynamic co-adaptation game.

1 Question

An implant (a device, a material, a controller, or a cell therapy) and the body it is placed in adapt to each other. The body regulates and remodels; the implant changes its state according to a built-in mechanism. Both processes are goal-directed, and the outcome of their interplay is an equilibrium of the coupled adaptation. Whether this equilibrium is close to a desired *healthy* state depends on how the implant adapts. The design problem is therefore

$$\min_{\theta \in \Theta} \Phi(x^*(\theta)) \quad \text{s.t.} \quad x^*(\theta) \text{ is an equilibrium of the game between body and implant}(\theta), \quad (1)$$

where θ parametrizes the implant’s adaptation mechanism and Φ measures the distance of the co-adapted outcome from the healthy state. The designer acts once, before the interaction, on *how the implant adapts*; it does not adapt a controller to an unknown plant. We call this *control of adaptation*, in contrast to adaptive control. The project name uses “co-evolution”; since evolutionary biology reserves this term for reciprocal genetic change across generations [6], “co-adaptation” may be the more precise term for publications.

2 Relation to existing approaches

Two-learner problem and co-adaptation. In brain-machine and body-machine interfaces, user and decoder adapt simultaneously; this “tale of two learners” [18] has been addressed with co-

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adaptive decoders [7], by showing that decoder adaptation shapes neural plasticity [17], with co-adaptation frameworks [5], and by tuning decoder learning rates for estimation performance [14]. Closest to the present question, brain and decoder have been modeled as strategic agents with own costs whose gradient learning converges to Nash equilibria depending on the learning rates [15]; decoder learning rate and regularization were then designed to *shape* the co-adaptive outcome [16], and machine learning algorithms were shown to select among game-theoretic equilibria in experiments with humans, including steering humans to the machine’s optimum [4]. The idea of designing a machine’s adaptation to shape a joint equilibrium is therefore not new. The present project differs in (A) continuous-time dynamic games with state dynamics, horizons, and running costs instead of static quadratic games with gradient learners, (B) designing the implant’s objective and adaptation law, judged by a designer criterion that differs from the implant’s own objective, (C) biological implant–body interactions, and (D) equilibria computed by optimal-control best responses, which admit nonlinear dynamics and state and path restrictions.

Human-in-the-loop optimization. Device parameters are optimized online against a measured human objective, e.g., metabolic cost under exoskeleton assistance [23]. The human enters as a black-box objective, not as a modeled strategic agent, and no equilibrium is predicted.

Adaptive and dual control. Adaptive control adjusts a controller to a plant with unknown parameters [1]; dual control adds deliberate probing to learn while controlling [8, 21]. In both, the plant is not goal-directed. Here the plant (body) adapts toward its own goals, and the controller’s adaptation law is the design variable. Dual control becomes relevant when the body’s objective is unknown and the implant must learn it while steering.

Learning in games, opponent shaping, steering, teaching. Learning dynamics and their equilibria are the subject of learning in games [3, 11]. Opponent-learning awareness lets an agent shape the anticipated learning step of the other [10]; payments can steer no-regret learners to a desired equilibrium [22]; machine teaching designs the inputs of a single learner [24]. These approaches act during the learning process; the design considered here acts once on the implant’s preferences.

Incentive design and strategic delegation. Problem (1) is a bilevel problem related to Stackelberg and incentive design in dynamic games [2, 13, 19]. Its designer–implant–body structure is that of strategic delegation: a principal gains by giving its agent preferences that differ from its own, because the agent’s preferences act as a commitment in the agent’s game with a third party [9, 20]. The indirect evolutionary approach [12] has the same structure, with preferences selected by the material outcome of equilibrium play. The example below reproduces the delegation effect.

3 A minimal co-adaptation game

States $b(t)$ (body) and $p(t)$ (implant) are one-dimensional, e.g., a tissue property and the corresponding implant setting. Each player owns one control, its adaptation rate:

$$\begin{aligned} \dot{b} &= u_B (p - b), & u_B(t) &\in [0, 1], & b(0) &= 0, & (2) \\ \dot{p} &= u_P (b - p), & u_P(t) &\in [-1, 1], & p(0) &= 1. & (3) \end{aligned}$$

This is the coupled relaxation $\dot{b} = \alpha(p - b)$, $\dot{p} = \beta(b - p)$ with the adaptation rates α, β replaced by decisions of the players. The body can only move toward the implant; the implant can yield to the body ($u_P > 0$) or lead away from it ($u_P < 0$). The body starts in its diseased state $s = 0$;

μ	start (u_B, u_P)	conv.	Φ	$\int (b - h)^2$	$\int (p - b)^2$	$c_P \int u_P^2$	$\int u_B$	$b(T)$	$p(T)$
0	(0.3, 0)	yes	1.953	0.372	1.542	0.040	1.541	1.038	1.249
0	(0.1, 1)	yes	1.946	0.374	1.532	0.040	1.545	1.036	1.244
0.1	(0.3, 0)	yes	2.884	0.567	2.208	0.110	1.010	0.868	1.049
0.1	(0.1, 1)	yes	2.883	0.566	2.207	0.110	1.010	0.868	1.049
0.25	(0.3, 0)	no	3.903	2.259	1.525	0.119	0.693	0.562	0.689
0.25	(0.1, 1)	no	3.661	2.030	1.496	0.135	0.628	0.578	0.692
0.5	(0.3, 0)	no	5.321	4.084	1.057	0.180	0.372	0.370	0.442
0.5	(0.1, 1)	no	5.321	4.084	1.057	0.180	0.372	0.370	0.442
1	(0.3, 0)	yes	6.593	5.585	0.793	0.216	0.257	0.256	0.308
1	(0.1, 1)	yes	6.593	5.585	0.793	0.216	0.257	0.256	0.308
2	(0.3, 0)	yes	7.203	6.517	0.488	0.198	0.226	0.195	0.232
2	(0.1, 1)	yes	7.205	6.520	0.488	0.198	0.226	0.195	0.232

Table 1: Design scan. Columns: implant mismatch weight μ , initial policy, audit passed, designer criterion and its parts, total body adaptation, and terminal states.

the implant is set to the healthy state $h = 1$ at implantation. On the horizon $[0, T]$, $T = 10$, the players minimize

$$J_B = \int_0^T (p - b)^2 + w_s (b - s)^2 + c_B u_B^2 dt, \quad (4)$$

$$J_P^\mu = \int_0^T \mu (p - b)^2 + (b - h)^2 + c_P u_P^2 dt, \quad (5)$$

with $w_s = 0.2$, $c_B = c_P = 0.1$. The body dislikes the mismatch with the implant, is attached to its set point (homeostasis), and pays for plasticity. The implant’s objective is the design: the weight $\mu \geq 0$ on the mismatch. The designer values health and integration equally and accepts the implant’s effort,

$$\Phi = \int_0^T (p - b)^2 + (b - h)^2 + c_P u_P^2 dt, \quad (6)$$

so that $\mu = 1$ is the *sincere* design: the implant pursues exactly the designer’s objective. Each player has one state, one control, and one objective.

Numerical method. For each design $\mu \in \{0, 0.1, 0.25, 0.5, 1, 2\}$, an open-loop Nash equilibrium candidate is computed with CorleoneGame.jl (sequential damped best responses, each an optimal control problem in single shooting with Ipopt; damping 0.2; at most 300 rounds; 20 and then 40 piecewise-constant control intervals). A candidate is accepted when no player improves its objective by more than the relative tolerance $2 \cdot 10^{-5}$ in a deviation audit by local solves. Since the game may have several equilibria, each design is solved from two initial policies, $(u_B, u_P) = (0.3, 0)$ and $(0.1, 1)$. An independent Python prototype (best responses by L-BFGS-B on 20 intervals, RK4 integration) gave the same values for the designs it reached: $\Phi = 1.963$ (Python, 20 intervals) versus 1.953 (CorleoneGame, 40 intervals) for $\mu = 0$, and 3.970 versus 3.970 for $\mu = 0.5$ and 6.12 versus 6.05 for $\mu = 2$, both on 20 intervals (`checks/grid20_2026-10-05.txt`). Of the 12 solves, 8 satisfied the audit tolerance; the others reached the round limit and are reported but not certified. The code is `co-evolution/simple.jl`.

4 Results

Figure 1 and Table 1 summarize the scan.

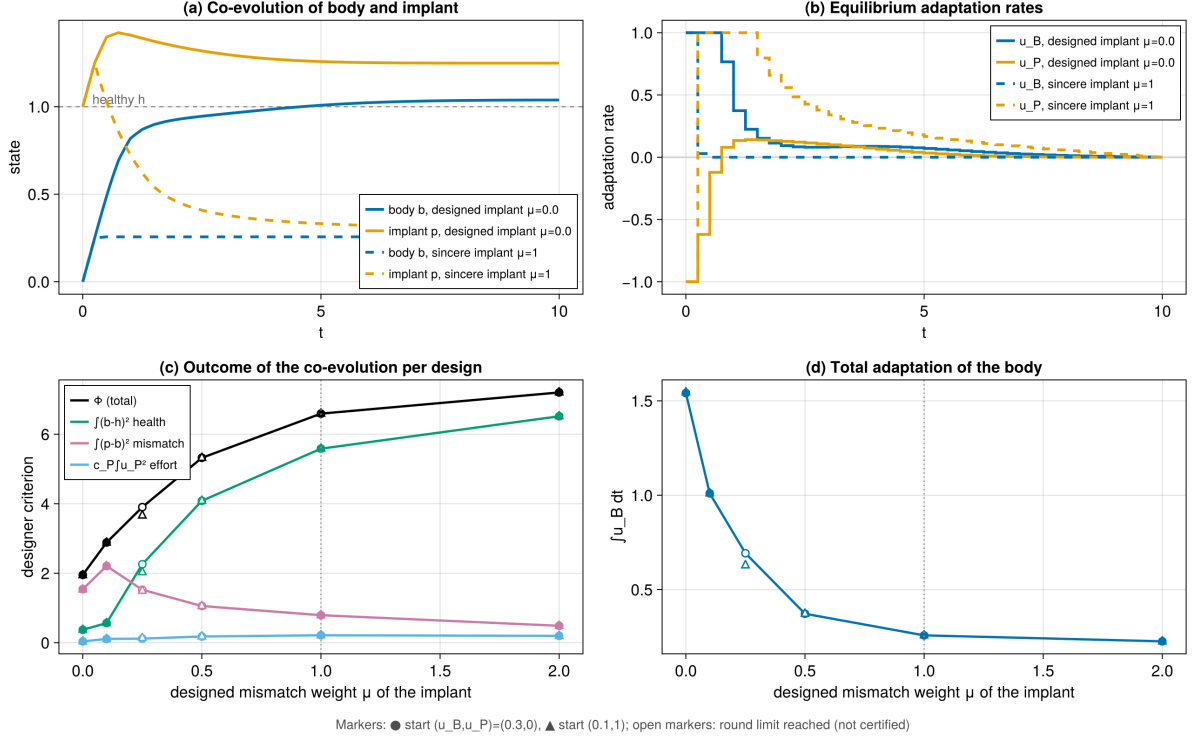


Figure 1: (a) Equilibrium trajectories of body b (blue) and implant p (orange) for the designed implant $\mu = 0$ (solid) and the sincere implant $\mu = 1$ (dashed); the gray line is the healthy state. (b) Equilibrium adaptation rates. (c) Designer criterion Φ and its parts over the design μ ; the dotted line marks the sincere design. (d) Total adaptation $\int u_B dt$ of the body. Lines connect the solves from the start $(0.3, 0)$; open markers indicate solves that reached the round limit.

The sincere implant is exploited. With $\mu = 1$, the body adapts in one short burst at the start ($u_B = 1$ on the first interval) and then stops, while the implant keeps yielding ($u_P > 0$) until the mismatch is small. In total the body adapts little ($\int u_B dt = 0.26$): it lets the implant do the adapting. Both meet far from health, $b(T) = 0.26$, $p(T) = 0.31$, and $\Phi = 6.59$, dominated by the health term (5.58).

The designed implant leads. With $\mu = 0$, the implant first leads away from the body ($u_P = -1$ and $u_P \approx -0.6$ on the first two intervals, p rises to about 1.4) and afterwards yields only slightly ($u_P \approx 0.1$). The body cannot reduce its mismatch cost except by adapting itself and does so strongly ($\int u_B dt = 1.54$), slightly beyond the healthy state, $b(T) = 1.04$, with the implant at $p(T) = 1.25$. The designer’s cost drops to $\Phi = 1.95$. The mismatch part is larger than with the sincere implant (1.54 versus 0.79), but health improves by far more (0.37 versus 5.58).

Interpretation. The co-adaptation is a game of who yields. Each agent prefers that the other adapts. An implant that cares about integration is predictably accommodating, and the body free-rides on this accommodation. An implant that does not care cannot be exploited; it commits to its position, and the body’s own mismatch aversion makes it follow. The designer, who does care about integration, achieves more of both health and overall value by *delegating* to an implant with different preferences. This is the commitment effect of strategic delegation [9, 20] in a dynamic co-adaptation game, and it shows “adaptive steering of another adaptive system”: the designer steers the body only through the implant’s adaptation mechanism.

Limitations. (A) The equilibria are open-loop and computed by local solves; the audit does not prove global Nash optimality. The two initial policies gave the same candidates up to small differences ($|\Delta\Phi| \leq 0.01$), except for the uncertified $\mu = 0.25$, but the control grid matters: on 20 intervals, the designs $\mu = 0.5$ and $\mu = 2$ reached certified candidates with $\Phi = 3.97$ and 6.05 , on 40 intervals uncertified 5.32 and certified 7.20 (Table 1). This indicates several equilibria or a grid-dependent equilibrium selection. All values for $\mu > 0$ remain far above Φ at $\mu = 0$, so the main conclusion does not depend on this. (B) Intermediate designs, notably $\mu = 0.25$, did not converge within the round limit; their values are not certified. (C) One parameter set; the optimum at the boundary $\mu = 0$ may change for other parameters, e.g., a stronger homeostatic attachment w_s . (D) The body’s objective is assumed known to the designer. (E) The model is a caricature without calibrated biology.

5 Next steps

The editable list is `co-evolution/nextsteps.md`. In brief: (A) consolidate the example (finer design grid, second design variable $p(0)$, robustness against a misspecified body, comparison with the Stackelberg design); (B) richer implant mechanisms: parametrized adaptation laws, feedback policies, and learning-rate designs as in [16]; (C) biologically grounded instances, e.g., bone remodeling with stress shielding, closed-loop neuromodulation, or tolerance constraints (rejection when the mismatch is too large); (D) unknown body: inverse games and dual control.

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